MUHAMAD IRFAN BIN MUHD RAMIZA

PIC TEST & CHARACTERISATION • PHOTONIC PACKAGING • INDUSTRIAL AUTOMATION
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Portfolio: ir02.github.io/Irfan-Portfolio/ | GitHub: github.com/IR02

PROFESSIONAL PROFILE

Experimental engineer pursuing an MSc in Optics and Photonics at Imperial College London after a 2:1 MEng in Mechanical Engineering. Developed and benchmarked test instrumentation, automated data acquisition and signal analysis, programmed Siemens PLC/HMI controls, and validated engineering simulations against experimental measurements. Building expertise in PIC testing, optical characterisation, fibre-to-chip integration and manufacturable photonic packaging for high-speed interconnects.

EDUCATION

Imperial College London | MSc Optics and Photonics Sep 2026 – Sep 2027 (expected)
Current study: Optical Measurement and Devices, Optical Communications, Optical Design, Fibres and Ultrafast Lasers.

University of Sheffield | MEng Mechanical Engineering (2:1) Sep 2021 – Jun 2026
Selected study: Industrial Automation (ELE426), Structural Vibration, Control Engineering, Finite Element Techniques, Numerical Methods, Robotics and Materials Integrity.

TECHNICAL SKILLS

Automation & controls: Siemens TIA Portal, PLC ladder logic, WinCC HMI / SCADA, sensor-actuator I/O, sequence control, timers/counters, conveyor and solenoid control, PID control fundamentals.
Instrumentation & measurement: PicoScope 4824A, IEPE accelerometers, piezoelectric sensors, Arduino DAQ, impact/modal testing, FFT and FRF analysis, digital image correlation (DIC).
Programming & engineering software: Python (NumPy, pandas, Matplotlib, scikit-learn), MATLAB, embedded C/ChibiOS, C++; Ansys Mechanical and Fluent, Fusion 360; Ansys Lumerical (familiarity).
Photonics knowledge: Silicon photonic ICs, waveguides, edge/grating couplers, optical modulators, Ge photodetectors, WDM links, optical loss budgets, fibre-to-chip integration and packaging (specialist training).
Manufacturing & laboratory: CNC/manual machining, precision valve-seat lapping, 3D printing, rig assembly, pneumatic and hydrostatic testing, test-repeatability assessment.

SELECTED ENGINEERING & RESEARCH PROJECTS

MEng Dissertation | Cryogenic CFRP Sensing and Test System 2025 – 2026
• Built and benchmarked a ~£26 piezoelectric-disc/Arduino sensing channel against an IEPE accelerometer/PicoScope reference (~£2,000–£4,000); both captured all five controlled pilot impacts.
• Manufactured 22 eight-ply CFRP coupons and constructed an Arduino-controlled, motorised liquid-nitrogen immersion rig; completed a five-minute pilot near 77 K.
• Processed signals with Python Hann-windowed FFTs; isolated a ~3.3 kHz response, compared 29% versus 43% peak-amplitude variability and diagnosed clipping/sample-rate limitations.

Industrial Automation (Sheffield) | PLC-Controlled Peg Sorting & HMI
• Programmed PLC ladder logic in Siemens TIA Portal for a Bytronic conveyor-based system sorting metal pegs and plastic rings; integrated sensor feedback, timed sequences and solenoid-controlled routing.
• Developed WinCC HMI/SCADA operator views for machine status, controls and live part counts, linking PLC inputs/outputs with material-handling and inspection stages.

AIM Photonics Datacom Design Challenge | Co-Packaged Optical I/O Jul 2026
• Contributed to conceptual design of a datacentre transceiver/package addressing a 1 Pb/s aggregate I/O target; evaluated WDM channel allocation, modulation and baud-rate scaling.
• Considered optical link budgets, fibre-to-chip coupling, propagation/connector losses, receiver sensitivity, bit-error-rate limits, energy per bit and package I/O density. (1 Pb/s was the brief target, not a measured result.)

Finite Element Analysis & DIC Validation | Clevis Assembly Mar – Apr 2025
• Modelled a 12.5 kN load case with quarter-symmetry in Ansys; reduced solver time 75% with <1% stress convergence at a 2 mm mesh size.
• Validated numerical strain against DIC (<1.5% X-direction discrepancy) and assessed peak von Mises stress of 217.94 MPa.

Experimental Modal Analysis | Steam Turbine Blades Nov – Dec 2024
• Performed instrumented impact tests on three blades at 10 locations (five impacts averaged per point); extracted first five modes in MATLAB and assessed resonance and damping behaviour.

Embedded Robotics | e-puck2 Autonomous Exploration & Tracking Feb – Mar 2025
• Developed a three-state search/approach/chase controller in embedded C (ChibiOS), combining proximity and time-of-flight sensors; achieved 100% in the assessed autonomous-exploration task.

Computational Fluid Dynamics | Backward-Facing Step Feb – Mar 2025
• Modelled turbulent separation and reattachment using Ansys Fluent with k-ω SST and near-wall inflation layers; compared numerical flow predictions against a reference benchmark (<2% reported difference).

Quantum Hackathon | QAOA Pathway Optimisation Mar 2025
• Mapped a simulated secure-facility layout to a weighted graph and implemented QAOA optimisation using Qiskit and Cirq; performed classical-parameter tuning and compared noiseless quantum-circuit outcomes.

INDUSTRIAL EXPERIENCE

RWNA Engineering Sdn. Bhd. | Engineering Technician, Malaysia Oct – Dec 2020
• Performed pneumatic/hydrostatic testing of industrial valves, precision grinding and seat lapping; dismantled, inspected and reassembled over 10 pressure-safety valves.
• Operated high-pressure water-jetting equipment and supported maintenance, diagnostics, test preparation and technical troubleshooting.

SPECIALIST PHOTONICS TRAINING

AIM Photonics Summer Academy | Hosted at MIT, Cambridge, USA Jul 2026
PIC fabrication and process design kits; DFM/process variation; optical testing, fibre attach, flip-chip/advanced packaging and datacentre interconnects.

ePIXfab Silicon Photonics Summer School | UPV, Valencia, Spain Jun 2026
Five-day programme on PIC modelling, passive/active devices, coupling, wafer-level characterisation, heterogeneous integration and packaging scale-up.

ePIXfab High-Speed Optical Modulators Workshop | Ghent University, Belgium Jun 2026
Silicon, InP and TFLN modulators; electro-optic/RF characterisation, high-speed interposers, packaging and heterogeneous integration.

LEADERSHIP, PROGRAMMES & AWARDS

University Leadership Development Programme (AXIATA) Jul – Aug 2025
• Selected among 73 of 807 applicants (~9%) for a 13-day leadership bootcamp; 2nd runner-up in the CEO Challenge pitching an AI-enabled equitable-education strategy to C-suite executives.

Sunway Group | Sunway Agility Challenge Aug 2025
• Completed 12-day live-case programme across strategy, property, healthcare and retail; presented acquisition synergies, competitor analysis and market strategies to senior leadership.

Co-founder | RWBI Foundation Jun 2024 – Present
• Co-founded a Malaysian STEM-access non-profit; led electrical-engineering workshops for 69 school students, achieving 84% satisfaction.

Head of UK Region | iCUBE International Apr – Sep 2022
• Managed seven departments, recruited 50+ officers and supported external partnership development and international project pitches.

Recognition:
• Mastercard Future Advisors Challenge 2026 — 1st Runner-Up;
• Sheffield Data & Design Hackathon — Most Creative Solution;
• Singapore and Asian School Math Olympiad — Bronze Medal.
• Ultramarathon Finisher — Sheffield to Leeds (~56KM, Elevation gain ~ 610m)
• Chess — Varsity

Languages: English (IELTS Band 8); Malay (native); German (CEFR B1+).